UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

«:> Hak Cipta Universiti Teknologi MARA

PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. A URL identifies the address of a web resource.
2. Interactivity level is irrelevant when choosing an authoring tool.
3. A hyperlink can link only to another web page, not to a file.
4. Card-based authoring tools are best for messages with a clear beginning and end.
5. HTML is used to structure content on web pages.
6. Kerning adjusts the space between individual character pairs.
7. A prototype is an early model of a product.
8. Streaming requires the whole file to download before playback.
9. Interlacing displays every line of a video frame at the same time.
10. Virtual reality replaces the real environment with a simulated one.
11. A bitmap image is made up of pixels.
12. RAM is a type of permanent storage.
13. Interactivity gives users control over content and flow.
14. A multimedia project always uses every media element.
15. A linear presentation allows users to choose their own path.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A. Matching options

A. Chroma key	B. Plotter	C. Multimedia
D. Sans serif	E. Plug and play	F. Graphics tablet
G. Keyframe	H. Storyboard	I. Serif
J. Scanner	K. Linear multimedia	L. Frame rate
M. Tweening	N. Card-based tool	O. Sampling

1. visual sequence plan
2. important frame in an animation
3. font with small strokes at character ends
4. integration of text, graphics, audio, video and animation
5. font without small strokes
6. frames displayed per second
7. input device that digitises documents
8. output device for large drawings
9. authoring tool organised like pages or cards
10. input device for drawing
11. presentation without user control of sequence
12. automatic device recognition
13. generating in-between frames
14. converting sound into digital values
15. replacing a colour background

C Hak Cipta Universiti Teknologi MARA

PART C (30 MARKS)

Choose the correct answer for each question.

1. Which image type scales without losing quality?
   A. Scan
   B. Vector
   C. Bitmap
   D. Photograph

2. Which colour model is subtractive?
   A. RGB
   B. CMYK
   C. HSL
   D. HSV

3. Convergence in multimedia refers to:
   A. Recording audio only
   B. Deleting media files
   C. Merging of media and computing
   D. Printing images

4. CMYK is used mainly for:
   A. Animation
   B. Screen display
   C. Sound mixing
   D. Printed output

5. Sampling converts:
   A. Images to vectors
   B. Analog sound to digital values
   C. Digital video to text
   D. Text to speech

6. Which is a benefit of multimedia in education?
   A. It removes all interaction
   B. It replaces all teachers
   C. It combines visuals and sound
   D. It uses only text

7. Which is an audio file format?
   A. MPEG
   B. HTML
   C. WAV
   D. JPEG

8. Which best defines multimedia?
   A. A printed book
   B. Text only
   C. Text, graphics, audio, video and animation
   D. A single image

9. A storyboard is:
   A. A visual sequence plan
   B. A colour model
   C. A video codec
   D. A sound file

10. Tweening creates:
    A. Video codecs
    B. Sound effects
    C. In-between frames
    D. Text fonts

11. MIDI stores:
    A. Images
    B. Musical notes and instructions
    C. Digitized audio samples
    D. Video frames

12. A bitmap image is made of:
    A. Sound samples
    B. Mathematical objects
    C. Pixels
    D. Text characters

13. A licence gives:
    A. Ownership of all works
    B. Permission to use a work
    C. Free hardware
    D. Unlimited bandwidth

14. Keyframes are:
A. Web links
B. Important frames in an animation
C. Audio samples
D. Colour profiles

15. Which format is commonly used for bitmap images?
A. HTML
B. MPEG
C. MIDI
D. PNG

16. Interactivity in multimedia means:
A. The user can control content and flow
B. The system plays automatically
C. Only text is shown
D. There is no navigation

17. Hypermedia is best described as:
A. A printed manual
B. Multimedia with hyperlinks
C. A single video file
D. Text-only media

18. Anti-aliasing is used to:
A. Increase file size only
B. Add sound
C. Compress video
D. Smooth jagged edges

19. A graphics tablet is used for:
A. Drawing input
B. Web browsing
C. Printing
D. Sound output

20. Amplitude of a sound wave relates to:
A. Colour
B. Loudness
C. Frame rate
D. Pitch

21. Target audience refers to:
A. Intended users
B. Software licence
C. Colour model
D. File format

22. MPEG is:
A. An image editor
B. A storage device
C. A font
D. A video compression family

23. HTML is used to:
A. Structure web content
B. Model 3D objects
C. Store audio samples
D. Compress video

24. RGB is used mainly for:
A. Video editing only
B. Printed output
C. Screen display
D. Audio recording

25. Which is a 3D animation task?
A. Sampling
B. Kerning
C. Modelling and rendering
D. Bookmarking

26. Morphing changes:
A. File format
B. Audio pitch
C. One shape into another
D. Font size

27. Cel animation is created by:
A. Drawing each frame by hand
B. Scanning text
C. Printing images
D. Recording sound

C Hak Cipta Universiti Teknologi MARA

28. A browser is used to:

A. Print documents only
B. Display web pages
C. Scan images
D. Edit audio

29. Which is an authoring tool category?

A. CMYK
B. JPEG
C. RGB
D. Card-based

30. Hertz measures:

A. Resolution
B. Frequency
C. Loudness
D. File size

PART D (40 MARKS)

QUESTION 1

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(4 marks)

QUESTION 2

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

(4 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

(4 marks)

QUESTION 3

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

(2 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

(5 marks)

QUESTION 4

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

(3 marks)

b) Describe TWO factors that affect image quality.

(3 marks)

c) Explain how colour model and resolution affect screen and print output.

(4 marks)

END OF QUESTION PAPER